Non-destructive testing – seeing inside materials

Non-destructive testing (NDT) is the art of finding cracks, pores and other flaws in a weld, an aircraft wheel or a pressure vessel without destroying the part. It is the backbone of safety on bridges, platforms, aircraft and power stations. The course brought together the physics behind the various methods – sound, electromagnetism, radiation – and their practical use. The page has an interactive ultrasonic A-scan and a figure on eddy currents.

> DTH, course 1845 (1987), 6 points: Non-destructive testing. The course was a natural continuation of Ultrasound from 1985. In Denmark the offshore industry in the North Sea was one of the big users of NDT in the 1980s.

Why test?

Structures rarely fail because the material is too weak. They fail because a small crack grows. A. A. Griffith showed in 1921 that a crack of length a in a brittle material becomes unstable when the stress reaches

$$\sigma_c = \frac{K_{Ic}}{Y\sqrt{\pi a}}$$

where K_{Ic} is the material's fracture toughness and Y a geometry factor close to 1. Halving the critical stress takes only a fourfold increase in crack length. Under alternating loads cracks grow gradually (fatigue), and Paris's law $da/dN = C(\Delta K)^m$ of 1963 describes the rate. The first jet airliners, the de Havilland Comet, crashed in 1954 because of fatigue cracks that started at the corners of cut-outs in the cabin; the platform Alexander L. Kielland capsized in the North Sea in 1980, killing 123 people, because a crack from a weld at a hydrophone fitting had grown through a brace. Non-destructive testing must find the cracks before they become critical – and fracture mechanics tells how small the cracks one must be able to find are, and how often one must look.

Surface methods

Visual inspection and dye penetrant

The most used method is still the eye, aided by magnifier, endoscope and good lighting. In penetrant testing a coloured or fluorescent liquid with low surface tension is applied and drawn into cracks by capillary action. The excess is wiped off, and a white developer draws the liquid out again, so that the crack shows as a clear line. Works on all non-porous materials, but only for cracks open to the surface.

Magnetic particles

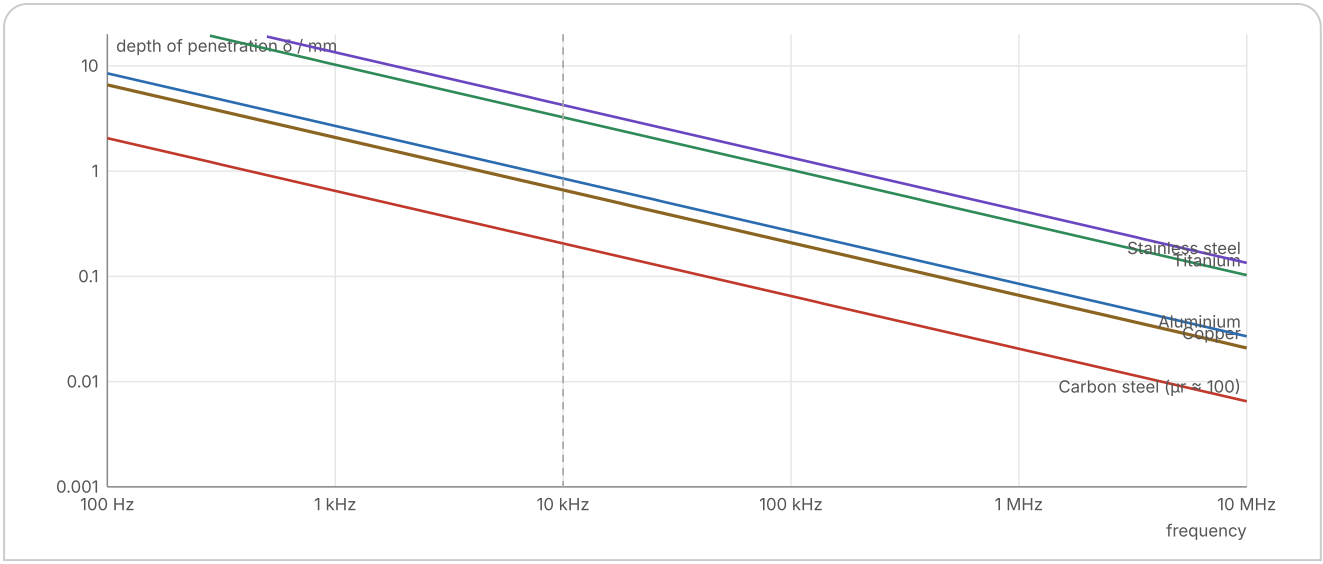
A ferromagnetic part is magnetised. Where a crack interrupts the field lines, the field "leaks" out of the surface, and fine iron powder – preferably fluorescent, in a liquid – gathers and outlines the crack. The method also finds cracks just below the surface, but only if they lie across the field; so one magnetises in two directions.

Eddy currents

A coil carrying alternating current is held against an electrically conducting part. The coil's field induces eddy currents in the part, which in turn change the coil's impedance. A crack forces the eddy currents to go round it, and the impedance changes – seen as a deflection in an impedance plane. The currents are concentrated near the surface and decay with depth as $e^{-z/\delta}$, where the standard depth of penetration is the skin depth (see field theory):

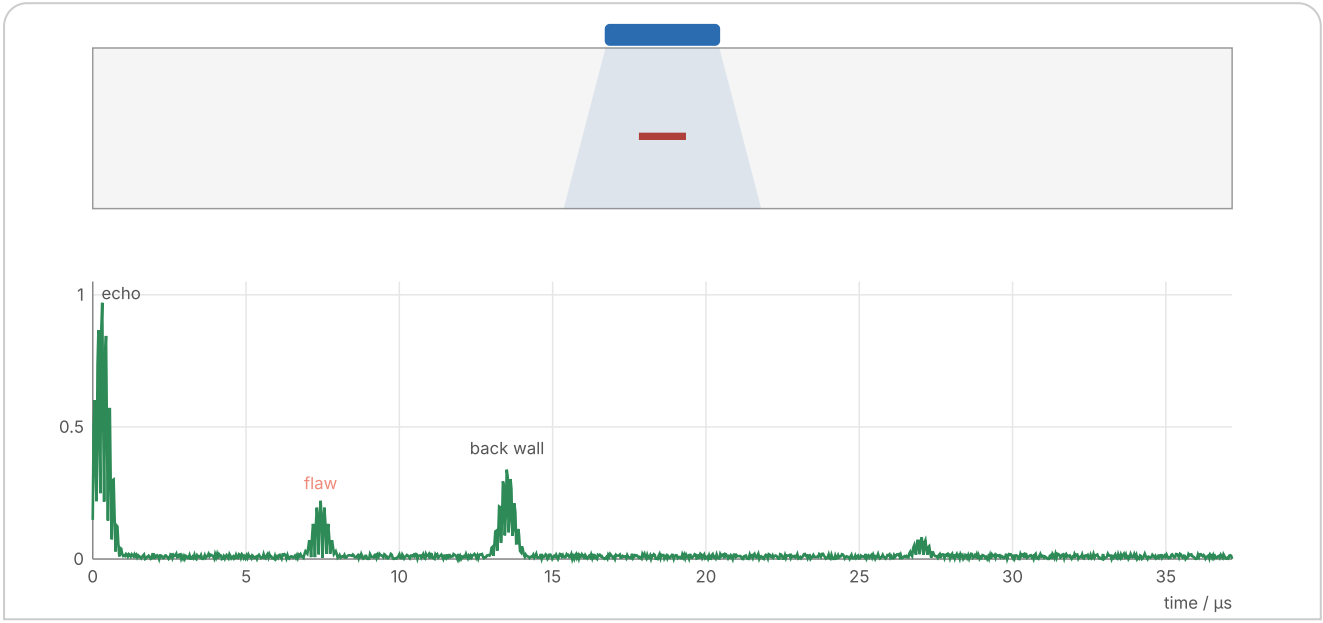
$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \mu_r \sigma}}$$

Low frequency reaches deep, high frequency gives high sensitivity to small surface cracks. In ferromagnetic steel μ_r is large, and the currents reach only very slightly below the surface. Eddy currents are used especially in the aircraft industry for rivet holes and wheels, and to check thousands of tubes in heat exchangers and steam generators.



Ultrasonic testing

Ultrasonic testing is the most important method for finding internal flaws in metal. A transducer (probe) of 1–10 MHz is coupled to the part with oil, gel or water and sends short pulses in. The screen shows the amplitude of the echoes against time – an *A-scan*. An echo from the back wall shows the thickness; an echo before the back wall is a flaw. The distance is found from the travel time, and the flaw size is estimated by comparing the amplitude with the echo from a calibration hole (the DGS method, from Krautkrämer: Distance–Gain–Size). Welds are inspected with angle probes that send shear waves obliquely, so that the sound hits the flanks of the weld. The near field and spreading of the sound field determine how well small flaws can be seen (see the sound field of a transducer).



Radiography

X-rays or gamma rays are sent through the part and form a shadow image on a film. The intensity decreases as $I = I_0 e^{-\mu x}$, so a cavity or pore, where there is less steel, gives more blackening. Wilhelm Röntgen took his first images in 1895, and already in the 1920s radiography was used to check castings and welds. In the field radioactive sources such as iridium-192 are used, which need no power. Radiography is good for volumetric flaws such as pores and slag inclusions, but poor for thin cracks not parallel to the beam – so it complements ultrasound. Radiation protection makes the method expensive and cumbersome.

Acoustic emission

When a crack grows, elastic energy is released in small bursts that spread as ultrasound through the structure. With many sensitive sensors spread over a pressure vessel, a bridge or a wind-turbine blade one can listen for the active cracks and locate them from the time differences – like an earthquake in miniature (see earthquakes). The method finds only flaws that are active under load, but monitors the whole structure at once.

How reliable is testing?

No method finds all flaws. Reliability is described by the *probability of detection* (POD) as a function of flaw size. It is determined by having many operators inspect test pieces with known flaws. A typical requirement in aviation is $a_{90/95}$: the flaw size found with 90 % probability, shown with 95 % confidence. The human factor is large – the same flaw is found by one operator and missed by the next. That is why operators are trained and certified to ISO 9712 at three levels.

Since 1989

- **Phased array and TFM:** Probes with 16–128 elements steer and focus the beam electronically; the Total Focusing Method gives sharp images of the whole cross-section.
- **TOFD** (time-of-flight diffraction) measures crack height precisely from diffraction at the crack tips and has become standard for welds.
- **Digital radiography and CT:** Film has been replaced by digital detectors, and industrial computed tomography gives 3D images of cast and 3D-printed parts.
- **Guided waves** are sent tens of metres along pipelines and rails and find corrosion from a single point.
- **Structural health monitoring** with permanent sensors on bridges, aircraft and wind turbines – and machine learning to interpret the signals.

Further reading

- Josef and Herbert Krautkrämer: *Ultrasonic Testing of Materials* (4th ed., Springer 1990)
- Peter J. Shull (ed.): *Nondestructive Evaluation: Theory, Techniques, and Applications* (CRC 2002)
- The Norwegian commission report on the Alexander L. Kielland accident, NOU 1981:11

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